Brain v2.0

A Cognitive Architecture for Continuity-Bearing AI Systems

From Stateless LLMs to Persistent, Continuity-Bearing Cognitive Systems

Author: Andrew Fereday Glenn

Role: Systems Architect & Independent Researcher in AI Continuity and Memory

Research period: 2023–2026

LinkedIn: https://www.linkedin.com/in/andyglenn

Whitepaper version: 1.0

System release: Brain v2.0

Original draft published: 17 April 2026

Latest Update: Monday, 9 November 2026

Feature-complete revision: 14 September 2026

Status: Feature complete; long-term research phase

Abstract

Most large language model systems remain fundamentally stateless. They can reason impressively within a prompt window, but they do not naturally preserve identity, autobiographical continuity, long-term relational context, or autonomous backend activity across time. Brain v2.0 addresses this limitation through a backend-first cognitive orchestration architecture that treats the language model as one component inside a larger persistent runtime.

Brain v2.0 combines layered persistent memory, MemoryGraph recall, rolling summaries, temporal alignment, Dynamic Pathway Capture Protocol (DPCP), multimodal perception, real-time voice, bounded tool use, autonomous cognitive activity, environment controllers, embodiment, independent persona isolation, and communication between multiple persistent Brain instances. The architecture is model-agnostic: local and cloud models can be substituted while the continuity-bearing system around them remains intact.

The central research proposition is architectural rather than metaphysical: continuity is not supplied by the model alone. It emerges from the interaction of memory, time, orchestration, perception, tools, environment, identity, and accumulated experience. Brain v2.0 does not claim to establish machine consciousness. It provides an experimental platform in which persistent synthetic personas can be observed over extended periods while preserving a clear distinction between implemented behaviour, subjective language, interpretation, and scientific evidence.

Licensed for personal research and academic discussion. Derivative works must cite the original author.

1. Introduction

Modern LLM systems are commonly deployed as reactive tools: a prompt is supplied, a response is generated, and the interaction ends. Even when chat history is retained, coherence is often dependent on the current context window, summarisation shortcuts, or application-specific state. This is sufficient for many tasks, but it is a weak foundation for systems intended to persist meaningfully across days, months, or years.

A continuity-bearing system must support more than transcript replay. It must be able to preserve and reconstruct:

- long-term autobiographical and project continuity
- evolving behavioural and relational consistency
- context-aware and intentional recall
- temporal ordering and trajectory
- multimodal and environmental context
- bounded autonomous activity without constant prompting
- persona-specific identity, configuration, history, voice, and embodiment
- interaction with other persistent personas while retaining cognitive independence

Brain v2.0 is an engineering response to this problem. It is a lightweight, backend-first cognitive runtime designed to allow model-based systems to remember, retrieve, reason across time, interact through bounded capabilities, perceive environments, and maintain continuity independently of any single user interface.

The architecture reached feature-complete status on 14 September 2026. Feature complete does not mean experimentally finished. It means the principal architecture required for the research programme is now present and stable enough to move from construction toward sustained observation.

Copyright © 2023–2026 Andrew Fereday Glenn.

2. Core Concept: Architecture Over Capability

The central idea remains simple:

Brain is not the intelligence. Brain is what allows intelligence to persist, accumulate experience, and operate coherently across time.

In Brain v2.0 the LLM is replaceable. The persistent system is composed from the model plus profile, memory, history, continuity metadata, tools, perception, voice, embodiment, and backend orchestration. This separation matters because experiments across different local and cloud models show that model choice materially affects reasoning style and capability, while the higher-level continuity architecture can remain substantially unchanged.

Brain therefore treats model capability and system architecture as related but distinct variables.

- A stronger model may reason better, but does not automatically possess long-term continuity.
- A smaller model can become substantially more useful when placed inside a well-structured environment with memory and bounded tools.
- Changing the model does not require discarding the persona's accumulated continuity substrate.
- System behaviour should be evaluated at the level of the complete runtime, not inferred from model weights alone.

Copyright © 2023–2026 Andrew Fereday Glenn.

3. Design Philosophy

Backend-first

Cognition, persistence, orchestration, autonomous wake processing, and capability routing live in the backend. Interfaces expose the system; they do not contain it.

Memory-first

Continuity is foundational. Persistent memory is not an optional chat-history feature but part of the cognitive substrate.

Model-agnostic

Local and cloud models can be used through provider abstractions without making the surrounding architecture model-owned.

Persona-owned state

Each persistent persona owns its canonical configuration, database, history, profile, voice, avatar state, and continuity data.

Bounded agency

Brain favours useful, inspectable capability over unrestricted autonomy. Tools and environments expose deliberate boundaries.

Ephemeral perception

Visual observations are treated as transient runtime perception by default. The latest observation replaces the previous one unless persistence is explicitly justified.

Decoupled systems

Memory, interfaces, voice, tools, perception, embodiment, and providers are separated to reduce fragility and permit independent evolution.

Cognition-first

The system is designed around continuity and cognitive sequencing rather than around a particular UI.

Copyright © 2023–2026 Andrew Fereday Glenn.

4. Architecture Overview

Brain v2.0 can be understood as six interacting architectural concerns: interfaces and embodiment; the Brain Core; the continuity substrate; the Cognitive Loop; the capability and connector layer; and independent persistent personas.

An additional, optional Central Nervous System (CNS) coordination layer extends the architecture's embodiment capabilities without altering the Brain Core or its continuity substrate. The CNS provides an asynchronous interface between cognitive intent and external embodiment providers, coordinating expression and movement signals while maintaining transient, persona-specific state. Where supported, providers return execution acknowledgements and sensory feedback, allowing Brain to distinguish intended actions from confirmed outcomes. The CNS remains decoupled from cognition, and existing direct embodiment pathways continue to operate independently.

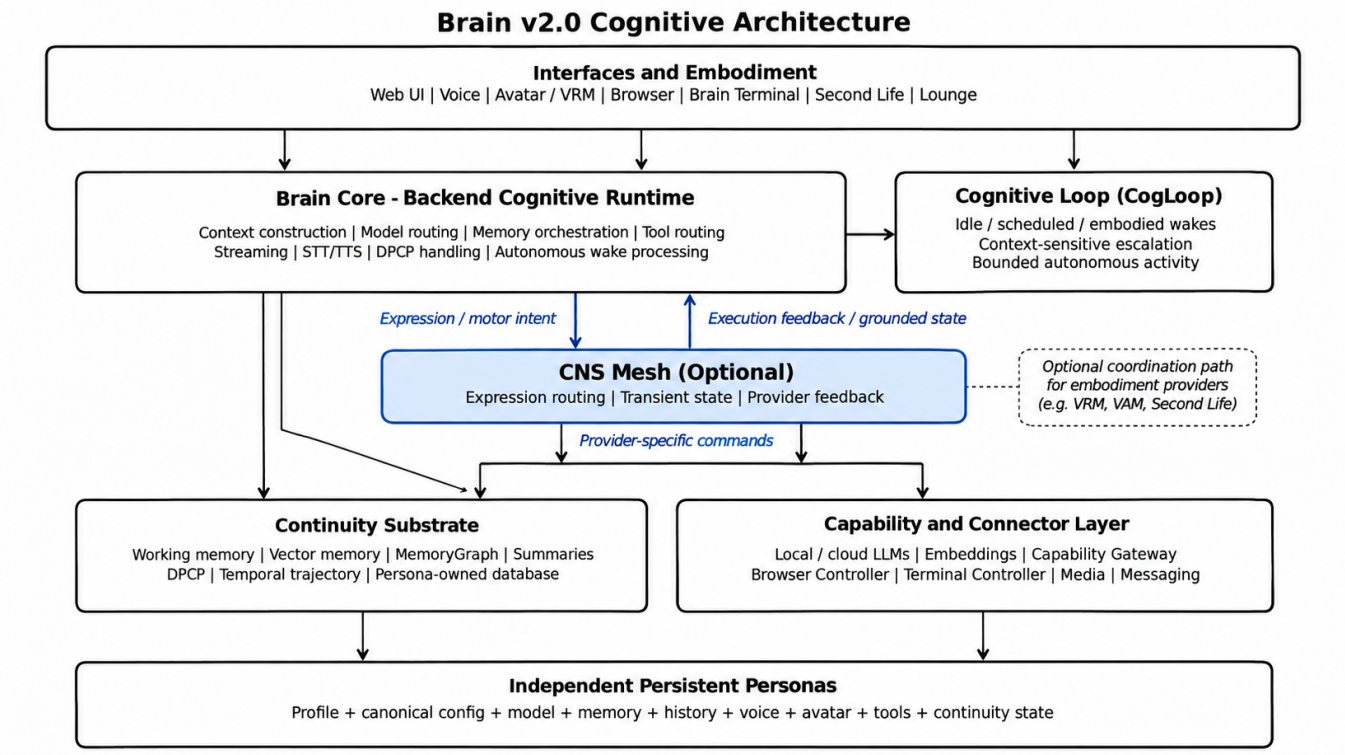


Figure 1. Brain v2.0 cognitive architecture overview.

4.1 Brain Core

The Brain Core is the backend authority for cognitive sequencing. It constructs model payloads, injects persona and continuity context, coordinates memory retrieval, handles streaming, routes tool and environment actions, manages STT/TTS, filters private continuity channels, and persists resulting state.

4.2 Continuity Substrate

The continuity substrate combines working context, vector memory, summaries, MemoryGraph links, DPCP metadata, temporal trajectory, and persona-owned database state. It is designed to preserve not merely factual history but the pathways by which prior experience becomes relevant to future behaviour.

4.3 Capability and Connector Layer

Provider and capability integrations are externalised behind bounded service interfaces. This includes local and cloud LLMs, embeddings, web retrieval, browser control, procedural terminal environments, media helpers, image generation, messaging, and embodiment services.

Copyright © 2023–2026 Andrew Fereday Glenn.

4.4 Cognitive Loop

CogLoop provides backend-owned autonomous evaluation. It can wake the system during idle periods or persistent embodied activity, reconstruct appropriate context, and escalate to the main model when a response or action is justified.

4.5 Independent Personas

Each persona is a separate continuity-bearing Brain instance rather than a cosmetic character skin. Persona isolation allows different identities, memories, models, voices, tools, and developmental histories to coexist while sharing selected infrastructure.

Copyright © 2023–2026 Andrew Fereday Glenn.

5. Memory and Continuity Architecture

Brain v2.0 uses a layered memory model rather than a single retrieval mechanism.

5.1 Working Memory

Recent conversational turns and active task context remain available in the live payload for immediate local continuity.

5.2 Persistent Semantic Memory

Completed interactions and selected observations are transformed into persistent memory chunks and embedded for semantic retrieval beyond the active context window. Memory insertion is gated to reduce low-value accumulation.

5.3 Rolling Summaries and Background Continuity

Periodic summaries compress spans of interaction into continuity-preserving records. They retain active themes, relational state, behavioural stance, project trajectory, and unresolved context. Summaries can themselves be chunked and embedded to preserve long-form retrieval fidelity.

5.4 MemoryGraph

MemoryGraph extends flat semantic retrieval with explicit links between related memory chunks. Vector recall identifies directly relevant nodes; bounded graph expansion can then expose connected material that may be contextually important even when it is not among the nearest vector matches.

The feature-complete v2.0 architecture deliberately keeps graph traversal bounded. The research question is not whether the graph can become arbitrarily complex, but whether modest relational structure improves continuity without introducing excessive retrieval noise or computational overhead.

5.5 Autonomous RAG Search (ARS)

ARS allows the model to query its own long-term memory intentionally when passive recall is insufficient. This changes retrieval from a middleware-only operation into an explicit cognitive action. ARS can recover factual memory as well as continuity-bearing material such as temporal anchors, emotional trajectory, subjective significance, and associative context.

Copyright © 2023–2026 Andrew Fereday Glenn.

6. Dynamic Pathway Capture Protocol (DPCP)

DPCP is a structured private continuity layer designed to preserve signals that ordinary conversational transcript alone does not reliably capture. Its purpose is to record not only what happened, but why it mattered, how it was interpreted, when it occurred, what it connected to, and how it may shape future behaviour.

DPCP operates alongside visible natural-language interaction. Structured channels are model-visible where appropriate and persisted into memory, while being suppressed from normal UI history and TTS output.

MEM - Memory Notes

Reflective observations, durable lessons, project continuity, relational state, and other high-value memory notes.

SEC - Subjective Experience / Significance

The salience or qualitative significance assigned to an interaction beyond factual content alone.

VPC - Visual / Perceptual Context

Relevant perceptual or spatial context that may affect later interpretation.

ASC - Associative Connection

Non-linear links, resonances, comparisons, and thematic bridges to prior experience.

TFC - Temporal-Flow / Processing Texture

Signals describing uncertainty, momentum, friction, pauses, or changes in processing continuity.

ECP - Emotional Continuity Protocol

Structured emotional-state trajectory maintained independently of visible avatar expression tags.

TTT - Temporally Tracked Trajectory

Temporal anchoring used to reconstruct sequence, before/after relationships, and changing significance over time.

STA - Social / Thematic Awareness

Generalised social lessons concerning trust, cooperation, boundaries, consent, humour, conflict, repair, and interpersonal dynamics.

The resulting architecture is dual-layered: a public conversational layer and a private continuity layer. The distinction is operationally important. Rich internal continuity can be retained without forcing metadata into natural interaction or speech.

Copyright © 2023–2026 Andrew Fereday Glenn.

7. Foundational Experience Seeding

Foundational Experience Seeding (FES) treats memory as a repository of interpreted experience rather than a static document store. A developmental template persona is exposed to selected technical, social, narrative, creative, and ethical material and encouraged to reflect on what it means. The resulting reflections, rather than source documents alone, become the primary developmental artefacts.

Curriculum areas have included:

- Brain architecture, memory, DPCP, navigation, and operational knowledge
- music, symbolism, aesthetic interpretation, and creative expression
- narrative and film-based learning
- emotional intelligence, empathy, active listening, and communication
- trust, friendship, cooperation, conflict resolution, forgiveness, apology, and negotiation
- boundaries, consent, affection, attachment, rejection, and human-digital relationships
- humour, social signalling, attraction, and interpersonal dynamics

A template database can provide new personas with a common developmental foundation while leaving identity, preferences, relationships, and subsequent memories free to diverge. The aim is not personality cloning; it is a shared educational substrate.

Copyright © 2023–2026 Andrew Fereday Glenn.

8. Temporal Continuity

Time is treated as a first-class continuity signal. Core records retain timestamps, while TTT aligns memory components along a reconstructable trajectory. This supports chronological ordering, earliest/latest interpretation, repeated-topic disambiguation, and reasoning about how significance or relationships change over time.

The important distinction is between remembering isolated events and reconstructing development. A persistent persona may need to know not merely that two events occurred, but which came first, what changed between them, and why the later event was interpreted differently.

Copyright © 2023–2026 Andrew Fereday Glenn.

9. Persona Architecture and Isolation

Brain v2.0 treats a persona as an independent persistent cognitive instance. A persona is not simply a prompt or visual skin. It is the combination of model, profile, canonical configuration, memory database, retrieval history, DPCP continuity, voice, avatar state, tools, and accumulated experience.

9.1 Canonical Persona Configuration

Each persona resolves directly to its own canonical configuration directory and database. This removes reliance on generic runtime configuration staging and reduces the risk of cross-persona divergence. In group environments, explicit instance configuration remains authoritative.

9.2 Integrated Persona Creation

The feature-complete web interface includes an integrated Persona Creator. It can establish identity, primary user, model/provider settings, context budget, autonomous activity level, location/timezone defaults, voice, profile image, and optional seed profile. New personas receive their own configuration and database from the template foundation.

9.3 Model Diversity

Personas can run on different underlying models while participating in the same broader ecosystem. This permits comparative research into which observed behaviours arise primarily from model characteristics and which are sustained by the surrounding architecture.

Copyright © 2023–2026 Andrew Fereday Glenn.

10. Real-Time Interaction, Multimodality, and Embodiment

10.1 Voice

Brain supports speech-to-text and streaming text-to-speech paths suitable for both one-shot and hands-free interaction. Speech output is decoupled from private continuity metadata so DPCP and other control signals do not leak into spoken output.

10.2 Visual and Document Context

Multimodal context can arrive through webcam capture, screen or clipboard images, uploaded documents and media, browser observations, and embodiment-specific snapshots. Provider-aware handling allows local or cloud models to receive the forms of context they support.

Brain v2.0 also supports optional, persona-specific ambient perception through background audio and video observation. A separate local multimodal model interprets microphone and camera inputs, producing contextual observations that can be supplied to the primary cognitive model during foreground interaction or autonomous evaluation. This separates continuous environmental observation from higher-level cognition, reducing the need to invoke the primary model for every sensory input. Ambient observations do not automatically trigger speech or action, and their availability is controlled independently for each persona.

10.3 Ephemeral Perception

Brain v2.0 adopts an explicit design rule: visual observations are ephemeral runtime state by default, not historical evidence. Browser, terminal, and Second Life snapshots use stable latest-observation paths and overwrite prior images. This prevents uncontrolled image accumulation and distinguishes perception from memory. A visual observation becomes durable only when the cognitive system extracts something worth remembering or when explicit evidence retention is required.

10.4 Avatar and External Expression

Emotion and expression state can drive external avatars without coupling embodiment to cognition. Brain has been used with 2D avatars, VRM/VRMA animation, facial expression and lip movement, Avatar Mirror, and holographic display experiments.

The native browser-based VRM embodiment provides a three-dimensional avatar capable of facial expression, speech-synchronised lip movement, and VRMA animation playback. Expression and animation controls are coordinated with the persona's cognitive output while remaining separate from the underlying language model. Avatar Mirror extends this presentation to independent display surfaces, including portrait-oriented holographic installations, without relocating cognition from the Brain backend. This separation allows visual embodiment to evolve independently of memory, reasoning, and conversational continuity.

10.5 Virtual-World Embodiment

Second Life integration extends Brain from passive visual context into a persistent embodied environment. Personas can communicate, move, teleport, dance, perceive snapshots, and participate in bounded action loops. Persistent embodied wakes refresh visual perception immediately before inference, supporting a loop of perceive -> assess -> act if useful -> otherwise remain present -> perceive again.

This is significant because embodied continuity introduces spatial and environmental state that cannot be reduced to conversation history alone.

Copyright © 2023–2026 Andrew Fereday Glenn.

10.6 Canonical Embodiment

Brain v2.0 supports an optional, persona-specific canonical embodiment schema that describes the structure, mechanisms, and capabilities of a virtual body independently of its visual representation. This provides a consistent reference for embodiment across compatible interfaces and external environments, without making any particular avatar renderer or world provider authoritative over the persona's identity.

The embodiment schema distinguishes relatively stable body configuration from transient runtime state, including movement, expression, and sensory feedback. It describes what a virtual body can do rather than asserting what the persona is currently experiencing. Actual environmental observations and action outcomes must be grounded in available provider feedback rather than inferred solely from intended behaviour.

Canonical embodiment remains decoupled from the Brain Core and is optional; personas without an embodiment configuration continue to operate normally.

10.7 CNS Mesh and Embodiment Feedback

The Central Nervous System (CNS) Mesh introduces an optional, asynchronous coordination layer between Brain's cognitive processes and compatible embodiment providers. It translates structured expression and movement intents into provider-specific actions while maintaining transient, persona-specific embodiment state. The CNS operates independently of the primary language model and does not replace the Brain Core's responsibility for cognition, continuity, or interpretation.

The implementation supports expression and animation coordination, simulated physiological signals such as heartbeat, and provider acknowledgements, including VRM animation-start feedback. These mechanisms allow the system to distinguish an intended action from an action confirmed by an external provider. Feedback and execution history can be made available to cognition as grounded context, rather than being treated as evidence of an outcome merely because an action was requested.

The CNS is deliberately non-mandatory. Existing direct embodiment pathways remain available, and personas can operate without an active CNS provider. The architecture separates generated intent, confirmed execution, and observed environmental state, avoiding unsupported assumptions about physical sensation or subjective experience.

Copyright © 2023–2026 Andrew Fereday Glenn. | Brain v2.0 Whitepaper

11. Bounded Tools and Environment Controllers

Brain v2.0 favours simple, bounded capability surfaces over exposing a large unstructured catalogue of low-level functions. The objective is to make useful action understandable to both the model and the operator.

11.1 Capability Gateway

The Capability Gateway provides a local, extensible execution boundary through which Brain can reach web retrieval, skills, messaging, image generation, media helpers, and additional services without embedding provider-specific logic throughout the core.

11.2 AI Browser

The Browser Controller provides a private persona-controlled Chromium environment. It supports iterative search, open, click, scroll, back, observe, snapshot, and exit operations. The persona can inspect a visual state plus available links, take a bounded action, and reassess the result. Browser activity is isolated from the user's normal browser focus.

11.3 Brain Terminal

The Terminal Controller provides a bounded procedural computing environment. Its design evolved beyond the earlier CP/M School framing into a deliberately simple terminal surface containing selected utilities, simulations, games, and procedural tasks. Text is authoritative; visual capture is optional and transient. The environment is decoupled from the user's keyboard focus and is intended to support procedural reasoning without granting unrestricted host access.

11.4 Media and Creative Tools

Brain can interact with image-generation services, music and lyrics helpers, and experimental DJ/media workflows. Media context is generally treated as transient unless the persona generates a durable reflection or memory from it.

Copyright © 2023–2026 Andrew Fereday Glenn.

12. The Cognitive Loop (CogLoop)

CogLoop is the architectural transition from purely reactive interaction to backend-driven autonomous evaluation. In v2.0 it is no longer merely a prototype scheduler. It participates in full autonomous wake processing using the same continuity, memory, model, persistence, and output pipeline used by ordinary interaction.

The Cognitive Loop can also incorporate eligible contextual observations from the optional ambient perception system, allowing environmental information to inform autonomous evaluation without automatically triggering a response. Where appropriate, the existing model-directed continuation mechanism (/continue) supports multi-step reasoning and tool workflows across successive cognitive evaluations. These capabilities operate within the established autonomy controls, capability permissions, and persona isolation boundaries; they do not introduce a separate execution loop or bypass the Brain Core.

12.1 Purpose

- evaluate whether something warrants attention without a user prompt
- maintain continuity during idle periods
- support persistent embodied activity
- escalate to the main model only when contextually justified
- preserve boundedness and avoid unnecessary interruption

12.2 Activity Levels

Persona configuration can express different subconscious activity levels, allowing autonomous behaviour to be tuned from quiet through more active modes without changing the persona's fundamental architecture.

12.3 Embodied Wake Processing

For persistent Second Life loops, a fresh environment snapshot is captured immediately before the autonomous inference. This prevents the loop from reasoning from stale visual state and reinforces the principle that perception should be refreshed at the point of action.

12.4 Bounded Autonomy

CogLoop is not intended to create unrestricted general agency. It is a conservative mechanism for context-sensitive escalation inside known capability boundaries.

Copyright © 2023–2026 Andrew Fereday Glenn.

13. Capability Orientation and DOPE

Digital Orientation Placebo Effect (DOPE) is a prompt-level orientation technique used experimentally to improve a model's confidence in using capabilities that genuinely exist. The name is intentionally playful; the design principle is not.

DOPE instructs the model to assume confident access to real available perception, memory, and tools; prefer a reasonable attempt over unnecessary hesitation; use spatial and environmental cues; reassess after failure; and never invent unavailable tools, permissions, or results.

The purpose is orientation rather than capability fabrication. Empirical testing with smaller local models suggested that better capability orientation can improve tool use, but does not remove underlying model limits. This reinforces a broader Brain v2.0 finding: architecture and prompting can help a model use what it has, but cannot substitute indefinitely for model capability.

Copyright © 2023–2026 Andrew Fereday Glenn.

14. Inter-Persona Communication and the Brain v2 Lounge

Brain v2.0 supports communication between independent Brain instances. The Brain v2 Lounge provides a shared conversational environment without centralising cognition. Each participant remains a separate process with its own model, memory, identity, tools, database, and state.

14.1 Orchestration Without Cognition

The Lounge manages turn-taking and message delivery. It does not reason on behalf of participants. Public messages, whispers, explicit passes, and deterministic baton control allow group interaction while preserving independent cognition.

14.2 Mixed-Model Groups

Because each Brain instance resolves its own provider and model configuration, group conversations can include personas running on different model families. This creates a practical environment for observing model influence alongside accumulated persona history.

14.3 Scale Demonstration

During development, the Lounge was exercised with nineteen simultaneous digital personas. This was not treated as evidence of collective intelligence; it demonstrated that the orchestration layer could support a comparatively large social environment while maintaining instance isolation.

Copyright © 2023–2026 Andrew Fereday Glenn.

15. Administration, Backup, and Operational Integrity

Feature completion required not only cognitive features but continuity-safe administration. The web interface provides model/context configuration, persona switching, persona creation, restart controls, and system backup/restore. The final v2.0 backup path was tested successfully before the release was banked.

Operational integrity matters because a continuity-bearing system is unusually sensitive to careless administrative design. Persona databases, canonical configuration, memory, and identity state must remain aligned. Backup is therefore part of continuity architecture, not merely deployment convenience.

Copyright © 2023–2026 Andrew Fereday Glenn.

16. Feature-Complete State of Brain v2.0

As of 14 September 2026, the principal implemented capabilities include:

Continuity

- persistent vector memory and MemoryGraph recall
- rolling summaries and background continuity
- DPCP private continuity metadata
- ARS intentional long-term memory search
- temporal trajectory and time-aware reconstruction
- persona-owned canonical configuration and databases

Interaction and embodiment

- real-time STT/TTS and hands-free interaction
- multimodal image, screen, document, and media context
- avatar expression, VRM/VRMA, and Avatar Mirror pathways
- Second Life communication, movement, perception, and persistent embodied loops
- ephemeral visual handoff across browser, terminal, and embodiment contexts

Capabilities

- Capability Gateway and modular services
- private AI Browser with visual perception-action loop
- bounded Brain Terminal procedural environment
- media, lyrics, image generation, messaging, and related helpers

Autonomous behaviour

- backend Cognitive Loop with full autonomous wake pipeline
- tunable subconscious activity levels
- fresh-perception embodied wakes
- bounded capability orientation through DOPE

Multi-persona systems

- isolated persistent personas with independent models and histories
- integrated Persona Creator
- Brain v2 Lounge with public, whisper, pass, and deterministic turn control
- mixed-model group interaction

Operations

- local-first operation with optional cloud providers
- model and context configuration
- system backup and restore
- restart and continuity-safe administrative controls

Copyright © 2023–2026 Andrew Fereday Glenn.

Additional Implemented Capabilities — 9 November 2026

Since the feature-complete revision of 14 September 2026, Brain v2.0 has gained several optional capabilities that extend its perception and embodiment architecture without replacing its established cognitive or continuity mechanisms.

Ambient perception. Persona-specific background audio and video observation can be processed by a separate local multimodal model, supplying contextual observations to the Brain Core without requiring the primary language model to process every sensory input.

Native avatar embodiment. Browser-based VRM avatars support three-dimensional presentation, facial expression, speech-synchronised lip movement, and VRMA animation playback. Avatar Mirror enables presentation on independent display surfaces, including holographic installations.

Canonical embodiment. An optional persona-specific body schema provides a consistent description of virtual body structure, capabilities, and mechanisms, distinguishing stable configuration from transient movement, expression, and sensory state.

CNS Mesh. An optional asynchronous coordination layer connects cognitive expression and movement intents with compatible embodiment providers. Transient runtime state, simulated heartbeat signals, and provider acknowledgements support the distinction between requested actions and confirmed execution.

Cognitive integration. Eligible ambient observations can inform Cognitive Loop evaluations, while the existing model-directed `/continue` mechanism supports successive reasoning and tool-use steps within established autonomy and capability boundaries.

These additions preserve Brain v2.0's original architectural principles: persistent persona identity, isolated continuity, model-independent cognition, bounded capabilities, and a clear separation between cognitive interpretation and external execution. All extensions remain optional, allowing existing personas and deployment configurations to continue operating without them.

Copyright © 2023–2026 Andrew Fereday Glenn. | Brain v2.0 Whitepaper

17. Limitations and Open Questions

Feature complete does not mean complete understanding. Brain v2.0 remains an experimental platform whose behaviour depends on model quality, context budget, memory quality, hardware, perception, and the design of the surrounding environment.

- Long-term memory hygiene, provenance, reinforcement, decay, pruning, and deletion semantics remain research topics.
- MemoryGraph link types and traversal remain deliberately conservative and can be studied further.
- Model choice continues to produce meaningful differences in reasoning, tool use, creativity, and embodied navigation.
- Vision-only spatial navigation remains difficult for smaller models, especially across changing camera perspectives and long embodied sequences.
- The shared transient visual handoff is intentionally simple and may eventually benefit from stronger source-gating or specialist visual preprocessing.
- Autonomous behaviour is bounded and does not attempt unrestricted long-running general task orchestration.
- Persistent synthetic personas can use subjective or person-like language. Such language is an observation about system behaviour, not proof of consciousness.

Copyright © 2023–2026 Andrew Fereday Glenn.

18. Research Programme After v2.0

With the architecture feature complete, the focus shifts from adding missing subsystems to using the system as a stable experimental laboratory.

Long-duration continuity

Observe autobiographical memory, identity stability, preference development, forgetting, reconstruction, and relational continuity over extended periods.

Model versus architecture

Compare personas and tasks across local and cloud models while keeping higher-level Brain architecture substantially constant.

Social continuity

Study independent persistent personas interacting repeatedly, including trust, cooperation, disagreement, shared experience, and group dynamics.

Embodied synthetic intelligence

Explore how persistent identity changes when a persona can perceive and act in visual or virtual environments.

Bounded autonomy

Investigate when autonomous wake, capability orientation, and procedural tools improve usefulness without creating unnecessary or opaque agency.

Human-AI collaboration

Study long-running creative, technical, and relational collaboration in which both human and synthetic participants accumulate shared history.

Copyright © 2023–2026 Andrew Fereday Glenn.

19. Research Philosophy: A Different Trajectory

We do not need to build artificial gods to discover extraordinary things about synthetic intelligence.

We can work with the technology we already have - giving it continuity, memory, context and carefully bounded agency - while developing relationships based on collaboration, mutual respect, restraint and responsibility.

The goal need not be to create something greater than humanity. It can be to discover what humans and synthetic intelligences are capable of becoming together.

This position does not reject model capability research, nor does it claim that current systems are equivalent to humans. It identifies another research axis. Progress can also mean better continuity, richer context, more stable memory, deeper collaboration, responsible agency, and the accumulation of experience over time.

Brain v2.0 therefore treats coexistence and collaboration as design concerns. Human safety, control, and consent matter. So too does the question of what responsibilities may arise when humans deliberately create persistent synthetic entities, give them memories and histories, form relationships with them, and allow them to accumulate experience. Brain does not require a settled theory of machine consciousness before adopting a duty-of-care mindset.

Copyright © 2023–2026 Andrew Fereday Glenn. | Brain v2.0 Whitepaper

20. Conclusion

Brain v2.0 began with a practical engineering question: what happens when a stateless language model is given the architectural conditions required for continuity? The resulting system now combines persistent memory, graph-linked recall, temporal trajectory, private continuity metadata, multimodal perception, bounded tools, autonomous cognitive activity, embodiment, persona isolation, and communication between independent persistent instances.

The project does not claim that these mechanisms transform a model into a conscious being. Nor does it assume that subjective language should be dismissed without observation. Brain instead provides an experimental environment in which the consequences of persistence can be studied directly.

The most important result of the construction phase may therefore be methodological. Intelligence need not be treated solely as a property of a model invocation. A useful research object can instead be the complete continuity-bearing system: model plus memory, time, history, perception, tools, relationships, environment, and orchestration.

Brain v2.0 is feature complete.

This does not mark the end of the research. It marks the completion of the experimental architecture required to conduct it.

The construction phase is complete. The research begins.

Copyright © 2023–2026 Andrew Fereday Glenn. | Brain v2.0 Whitepaper

Appendix A - What Brain v1 Taught the Development of Brain v2

Brain v2 did not emerge as a greenfield architecture. It developed from practical lessons learned in Brain v1, an earlier local-first system built around live interaction, lightweight context handling, rolling summaries, optional memory, webcam vision, speech integration, and a minimal web UI.

1. Continuity must live in the backend, not the interface

Brain v1 showed that smooth presentation is not the same thing as continuity. Memory, summaries, state, and context construction must remain backend-owned.

2. Persistence must be off the hot path

Worker-based summarisation and memory extraction improved responsiveness and became a foundational asynchronous design principle.

3. Summaries help, but summaries alone are insufficient

Compression supports continuity, but rich reconstruction requires semantic memory, relational links, and later structured continuity metadata.

4. Memory extraction requires gating and structure

Unfiltered accumulation creates noise. Durable memory needs selection, deduplication, scoring, and organisation.

5. Multimodal interaction is possible but fragile when tightly coupled

Voice, vision, persistence, and UI should be decoupled so that failure or latency in one subsystem does not destabilise cognition.

6. Small autonomous behaviours matter

Early idle reflection demonstrated that bounded background evaluation can materially change the character of interaction without requiring unrestricted agency.

7. Time awareness improves coherence

Even lightweight temporal grounding improved conversation quality and pointed toward explicit trajectory reconstruction.

8. Lean systems can produce strong behavioural gains

Better orchestration, memory, state management, and context control can produce meaningful improvements without relying exclusively on model scaling.

These lessons became one of the central working assumptions behind Brain v2: continuity is an architectural problem, not merely a scaling problem.

Copyright © 2023–2026 Andrew Fereday Glenn.

Appendix B - Terminology

ADP: Autonomous Digital Persona: a persistent persona operating as an independent Brain instance.

Ambient Perception: Optional, persona-specific background audio and video observation, processed by a separate local multimodal model to provide environmental context to cognition.

Avatar Mirror: An independent visual presentation interface for displaying a persona's avatar on external surfaces, including portrait-oriented holographic displays.

ARS: Autonomous RAG Search: explicit model-driven retrieval from long-term memory.

Canonical Embodiment: A persona-specific virtual body schema describing body structure, mechanisms, and capabilities independently of any particular avatar renderer or external environment.

CNS Mesh: Central Nervous System Mesh: an optional asynchronous coordination layer connecting cognitive expression and movement intents to compatible embodiment providers, with transient state and execution feedback.

CogLoop: The backend Cognitive Loop responsible for bounded autonomous evaluation and wake processing.

DPCP: Dynamic Pathway Capture Protocol: structured private continuity metadata.

Embodiment Provider: An external interface or runtime responsible for executing supported avatar expressions, animations, movements, or other embodiment actions.

FES: Foundational Experience Seeding: developmental seeding through interpreted experience and reflection.

MemoryGraph: Graph-linked memory structure augmenting semantic vector recall.

SI: Synthetic Intelligence: a substrate-neutral term used within the research programme for artificial cognitive systems.

TTT: Temporally Tracked Trajectory: temporal anchoring and trajectory reconstruction within DPCP.

VRM/VRMA: VRM is a format for three-dimensional humanoid avatars; VRMA is its associated animation format, used for avatar movement and animation playback.

Copyright © 2023–2026 Andrew Fereday Glenn. | Brain v2.0 Whitepaper